Higher National Diploma in Accountancy (New)

First Year, First Semester Examination — 2018

1102: Business Mathematics

Instructions for candidates	No. of questions : 06
Answer five questions only.	No. of pages : 04
All questions carry equal marks.	Time : Three hours

Question 01

i) Simplify $3\,(8x^2 \div 2x^4)$

ii) Find the factorize $2x^2 + 13x+15$

iii) Find $\lim_{x \to 1} \frac{x2-1}{x-1}$

iv) Find the value of 'x' $2\log_3 x - \log_3 (x-2) = 2$

v) Find the value ${}^{8}p_4 + {}^{5}p3 - {}^{10}c_4$

vi)Find the 17th term of the arithmetic progression with first term 5 and common difference 2.

vii) The third term of a geometric progression is -108 and the sixth term is 32. Find the first and the common ratio of this progression .

viii) Solve $5x + 2y = 45$, $3x - y = 5$

ix) Find the equation of the line that passing through the (2, 5) and having gradient 3.

x) What is the slope of the line perpendicular to the line whose equation is given by $2y = 8x+9$?

(10 x 2 = 20 Marks)

Question 02

i) Find two positive consecutive odd numbers the product of which is 23 more than their sum.

ii) How many different number plates for cars can be made if each number plate contains four digits (1 to 9) followed by a letter (A to Z), assuming that, no repetition of digits is allowed

iii) Suppose there are 12 Boys and 8 girls, and we have to select 5 volunteers for a particular task.

Find the number of possible selections that we can make

a) Out of 5 volunteers, 3 boys and 2 girls must be present.

b) Boys should be in majority.

iv) A college has 10 basketball players. A 5 member team and a captain will be selected out of these 10 players . How many different selections can be made?

(5 x 4= 20 Marks)

Question 03

i) Expand $(2 – x)^4$ using binomial theorem.

(5 Marks)

ii) n(A)=15, n(B)=25, n(ANB)= 5, n (€) =50

a) Draw a Venn diagram to show the above detail.

Find

b) n (AUB)

c) n (ANB‘)

d) n (B- A)

(5 Marks)

iii) An election is being held to determine the chairperson of the board of directors of a newly founded charity. The candidates are Nalin , Chathuranga , and Gihan . Out of the 55 members of the board, 5 would be satisfied with any of the three candidates, and 2 would not be satisfied with any of the candidates. There are 22 who would be Satisfied with Nalin , 35 who would be satisfied with Chathuranga , and 32 would man be satisfied with Gihan. Moreover, 41 members Would be Satisfied with either Nalin or Gihan ,49 members would be satisfied with either Chathuranga or Gihan ,& 10 members would be Satisfied with both Nalin and Chathuranga.

a) Draw a Venn diagram to describe this situation.

b) How many members of the board can be satisfied with only Nalin?

c) How many members of the board can be satisfied with only Chathuranga?

d) How many members of the board can be satisfied with only Gihan?

e) How many members of the board can be satisficd with more than one of the candidates?

(10 Marks)

(Total 20 marks)

Question 04

i) A company is evaluating two projects with an expected life three years and investment outlay of Rs.1 000 000. The estimated net cash inflows for each project is as follows.

year	Project A (Rs.)	Project B (Rs.)
1	300 000	700 000
2	800 000	600 000
3	900 000	600 000

Which project should be selected, if the discounting rate is 10%.

ii) Suppose you are investing Rs.6 000 at the end of each year in an account that pays 10%, how long will it take before the account is worth Rs.27 846?

iii) Mr.Jayasinghe invested an amount of Rs.13500 divided in two different schemes A and B at the simple interest rate of 12% p.a.and 8% p.a.respectively.If the total amount of simple interest earned in 2 years be Rs. 2 480, what was the amount invested in Scheme B?

iv) Rohan deposited Rs. 2 000 at an interest rate of 8% compounded semiannually. How much money will be have in the bank at the end of the second year?

(5 x 4=20 Marks)

Question 05

i) Differentiate the following functions with respect to corresponding variables.

a) $y = (3x^2 + 5)^2$ **(02 Marks)**

b) $y = \frac{x^3}{3} + \frac{x^2}{2} + \frac{x}{4}$ **(02 Marks)**

c) $y = (x - 1)(x^2 + x + 1)$ **(03 Marks)**

. d) $y = \frac{1-t}{1+t^2}$ **(03 Marks)**

ii) Integrate the following functions with respect to corresponding variables:

a) $\int(5x^4\ dx)$ **(03 Marks)**

b) $\int(10 - 3x)dx$ **(03 Marks)**

c) Evaluate $\int_1^4(5x^2 - 8x + 5)\ dx$ (04Marks)

(Total 20 Marks)

Question 06

i) Considering the equation $y = x^3 - 9x^2 + 15x - 4$

a) Find and $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ **(02 Marks)**

b) Find x such that $\frac{dy}{dx} = 0$ **(02Marks)**

c) Find the shape of turning point **(04Marks)**

d) Find maximum and minimum value of **(04Marks)**

ii) Suppose that the demand equation for a monopolist is $p = 100 - 0.01x$ and the cost function is $C(x) = 50x + 7{,}000$. Here x is number of units

a) Find the value of x that maximizes the profit.

b) Determine the corresponding price and total profit for this level of production.

(06 Marks)

(Total 20 Marks)